Uniqueness of the Classical Langton Ant Highway from Its Layer Profile

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Abstract

We prove a computer-assisted rigidity theorem for finite-support periodic highways of the classical two-colour Langton ant. Suppose a heading-restoring period has full drift $(2, -2)$, net black growth twelve, and transverse width ten. If each strictly interior transverse level contains at most one odd translation class, then the period is 104 and its read word is the standard highway. Neither the complete wake, the position of its missing interior strand, nor a period bound is prescribed. An analytical parity argument forces both boundary residues to be occupied and reduces the transverse profile to five possibilities. A residue-independent integer certificate supplies period bounds between 160 and 192. Exact modular edge counts reduce the remaining choices to 76 normalized wake assignments. Complete ordered-event searches leave only the standard trace. The classification derives the standard layer profile and its longitudinal residues from these sparse geometric data. General wake-dependent period bounds and a terminating inverse-synthesis algorithm for every non-axis drift are retained as a second part of the theory.

Keywords. Langton's ant; finite support; periodic highway; computer-assisted proof; wake rigidity; exact enumeration.

1 Introduction and main theorem

The classical Langton ant is a deterministic walker on a two-coloured square lattice. On white it turns right, on black it turns left; it then toggles the current colour and moves one lattice edge. The finite-support Highway Conjecture asserts that every finite initial black set eventually produces the standard period-104 translating trace. This statement includes both an *entrance problem*, concerning eventual periodicity, and a *classification problem*, concerning the possible periodic limits. The present paper advances the second problem.

A periodic highway emits an unbounded black wake, even though its seed is finite. The wake can be specified by the translation classes receiving an odd number of visits in one period. A previous exact classification fixed all these classes to those of the standard highway [6]. Here the classes themselves are unknown. We retain only a small set of numerical and sparsity hypotheses and derive the complete standard wake as a conclusion.

Definition 1.1. For a heading-restoring word $w = w_0 \cdots w_{P-1}$, let $q_i \in \mathbb{Z}^2$ be its pre-update positions and $d = q_P - q_0$ its drift. For $d = (2, -2)$ the full classes modulo $\mathbb{Z}d$ have labels

$$(t, r) = (x + y, x \bmod 2).$$

Let o_t be the number of classes at level t visited an odd number of times in the period. Let $g = \#\mathrm{R} - \#\mathrm{L}$ and

$$W = \max_i(q_{i,x} + q_{i,y}) - \min_i(q_{i,x} + q_{i,y}).$$

The wake is *interior-thin* when $o_t \le 1$ at each level strictly between the two extrema.

The word "thin" refers to odd-class occupancy, not to the number of visits or crossings. There may be many visits on an interior level. Nor does it say that an ordinary row of the infinite colouring contains at most one black cell.

Theorem 1.2 (Thin-wake rigidity)**.** *Let a word w repeat forever under the classical Langton rule from a complete finite black seed. Suppose it restores the heading, has full drift $(2, -2)$, has net black growth $g = 12$, and has transverse width $W = 10$. If its wake is interior-thin, then $P = 104$ and w is a cyclic phase of the standard highway.*

There is no period bound in this hypothesis. The seed need not occupy a prescribed box, and genuinely unread stationary debris is allowed. The theorem also applies to an eventual periodic suffix after an arbitrary finite transient. Its boundary state is again a complete finite-support state.

Corollary 1.3. *Any nonstandard finite-support periodic highway with drift $(2, -2)$, growth twelve, and transverse width ten has both longitudinal residue classes odd on at least one strictly interior transverse level.*

These numerical and structural hypotheses are not asserted for every finite seed. In particular, Theorem 1.2 does not solve the entrance problem or classify other drifts, growths, widths or thicker wakes. Repeating the standard period twice changes the drift to $(4, -4)$ and growth to 24; it is not a second period in the fixed-drift problem.

The proof combines analytical reductions with finite integer computations. The supplied-word realization criterion is prior work [2]; short-factor constraints belong to the symbolic study of the ant [1]. The specific upper certificate and ordered-event engine are retained from the standard-wake project [6]. The contribution here is the removal of the complete-wake and missing-level hypotheses, together with the complete classification of the resulting finite family. No lower-width exclusion is used as a premise.

2 Finite support and stabilized visit order

A complete state is $X = (B, p, h)$, where $B \subset \mathbb{Z}^2$ is finite and $h \in \{N, E, S, W\}$. Write F for one literal update and $\chi(x, y) = (-1)^{x+y}$. The quantity

$$\sigma(X) = \begin{cases} \chi(p), & h = N, S, \\ -\chi(p), & h = E, W \end{cases}$$

is invariant: each update changes both position parity and heading axis. All times below are pre-update times.

Fix a nonzero drift and write the positions in one class as $q_i = b + a_i d$. A visit to the physical cell $b + nd$ at phase i occurs, when present, at time

$$(n - a_i)P + i.$$

Thus the stabilized order of phases in that class is decreasing a_i, followed by increasing i. Equivalently, its keys are $i - Pa_i$.

Lemma 2.1 (Supplied-word criterion)**.** *A heading-restoring nonzero-drift word admits a finite-support realization if and only if every class, in stabilized order, alternates* $\mathrm{R}, \mathrm{L}, \mathrm{R}, \mathrm{L}, \ldots$ *starting with* R*.*

Proof. Far enough along a class every phase is present, and infinitely many such physical cells are initially white. Their first turn is R; successive reads toggle the colour. This proves necessity. For sufficiency, the phases available at $b + nd$ form the suffix obtained by removing all phases with $a_i > n$. For $\min a_i \le n < \max a_i$, initially blacken this cell precisely when the number of removed phases is odd; leave every other cell white. Each available suffix starts with the correct colour and alternates. Induction in real time verifies every prescribed update. Only finitely many boundary cells have been blackened. This is the known construction in [2]. □

Each class therefore contributes zero or one to $\#\mathrm{R} - \#\mathrm{L}$, according as its cardinality is even or odd. In particular,

$$g = \sum_t o_t. \tag{1}$$

Lemma 2.2 (Extreme classes)**.** *Every visited class on a transverse extreme has exactly one event, a right turn. In the top-rail normalization the top is* $t = 0$*, the head faces east,* $\sigma = -1$*, and the bottom is* $t = -10$*.*

Proof. At an extreme, one neighbour on each coordinate axis is outside the trace. The invariant σ fixes the arrival axis. Hence every visit to a fixed extreme class has the same arrival and departure directions, and thus the same turn. Lemma 2.1 forces this constant stabilized word to be the singleton R. At the top the only compatible right-turning pose is east-facing; at the bottom it is west-facing. Translate a top event to the origin. The stated gauge and width follow. □

There are at most two extreme classes on either boundary, because the quotient is by the full vector $(2, -2)$, not its primitive direction. The geometric interval is a consequence of the candidate's stated width and normalization; it is not a wall added to the original dynamics.

3 Five possible profiles, derived analytically

Let m_t count the phase events on level t. Let u_t count straight passages from $t - 1$ through t to $t + 1$. Flow balance in the closed transverse walk gives the same number of reverse straight passages. At even t, right turns reverse the transverse direction and left turns continue it; at odd t the roles are reversed. Consequently

$$m_t = (-1)^t o_t + 4u_t. \tag{2}$$

The movement axis alternates, so there are $P/2$ horizontal moves. Their net displacement is two, implying that $P/2$ is even. Summing (2) gives

$$P \equiv 0 \pmod 4, \qquad \sum_t (-1)^t o_t \equiv 0 \pmod 4. \tag{3}$$

Proposition 3.1 (Profile reduction)**.** *Under Theorem 1.2's hypotheses both extreme levels have two odd classes. Exactly one interior level is empty, and it is one of*

$$h \in \{-9, -7, -5, -3, -1\}.$$

All other interior levels have one odd class.

Proof. The two rails supply at most four odd classes; the nine interior levels supply at most nine. Growth twelve forces the rail total to be three or four. If it is three, all nine interior levels are occupied. Their signed sum is $4 - 5 = -1$, so the total in (3) is two, a contradiction. Thus the rail total is four and exactly eight interior levels are occupied. If h is the empty level, the signed sum is $4 + (4 - 5) - (-1)^h = 3 - (-1)^h$. It is divisible by four exactly for odd h. □

For a fixed h, list the eight occupied interior levels in increasing order. An eight-bit mask assigns their longitudinal residues. There are 256 masks. Changing the origin to the other singleton top event swaps every residue, i.e. replaces the mask by its bitwise complement. We choose the origin so that the highest occupied interior strand has residue one, giving masks $128, \dots, 255$. This normalization uses an actual top event and a lattice translation; it uses neither reflection nor colour complementation.

Corollary 3.2 (Standard layer profile)**.** *Suppose the full drift is* $(2, -2)$ *and, after normalizing the highest odd level to zero, the odd-class counts at levels* $-10, \dots, 0$ *are*

$$(2, 1, 1, 1, 1, 1, 1, 0, 1, 1, 2),$$

with no odd classes elsewhere. Every finite-support periodic realization has period 104 *and the standard read word. The longitudinal residues on the eight singleton levels need not be specified.*

Proof. The supplied-word criterion gives growth twelve. Every visited extreme is odd, by Lemma 2.2, so the actual width is ten. The supplied profile is interior-thin. Apply Theorem 1.2. □

In particular, specifying all twelve standard class labels gives the complete-wake uniqueness theorem of version 0.2.0 [6]. The direct profile computation used 256 masks, of which 216 failed the count relaxation; complementary rephasing reduced the remaining forty to twenty. The present five-profile proof subsumes that classification while also deriving the missing level. Both proof narratives remain in the source package; the current full reproduction uses the stronger 76-case inventory.

4 An all-period upper certificate

The period is not a supplied search parameter. The following integer certificate bounds it before chronology is enumerated.

The retained necessary graph has states (t, h, a) for $-10 \le t \le 0$, with the required heading axis, where a is a suffix context avoiding 588 verified impossible factors of length at most 21. Every factor contains two consecutive reads of a physical cell with the same turn and is therefore impossible for any initial colouring. All placed contexts stay in the stated interval. The finite recipe gives 54,718 vertices and 90,158 edges. Every periodic physical trace lifts, after history warmup, to a closed walk in this graph. The walk need not be a simple cycle; the converse is not assumed.

Set

$$\alpha_t = -2(-1)^t(t + 5)^2, \tag{4}$$

whose values from -10 through zero are

$$(-50, 32, -18, 8, -2, 0, -2, 8, -18, 32, -50).$$

The integer recipe starts all vertex potentials at zero and repeatedly raises y_w to $\max(y_w, y_v + 1 + \alpha_t\varepsilon)$ along edges. A guard at 180 makes this reconstruction a finite procedure; every edge is checked afresh at termination. The retained certificate passes all inequalities

$$y_v - y_w + \alpha_t\varepsilon \le -1, \qquad \varepsilon = 1 \text{ (R)}, \quad -1 \text{ (L)}. \tag{5}$$

Its maximum potential is 180. No floating-point optimizer or solver infeasibility report is a premise of this certificate.

Lemma 4.1 (Profile budgets)**.** *The five profiles of Proposition 3.1 have respective all-period bounds* $192, 168, 160, 168, 192$.

Proof. Sum (5) along the actual lifted closed walk. Vertex potentials telescope. Lemma 2.1 gives the signed turn excess o_t at level t, so

$$P \le -\sum_t \alpha_t o_t.$$

Substitution of the five profiles yields the displayed bounds. □

The same graph and potentials certify all profiles. The graph forgets the longitudinal residue only for this necessary upper bound. The exact search in the next section retains it. Complete initial black fields are not deleted or replaced by a chosen template.

5 Finite residue and count reductions

At a class adjacent to the processed part, let I, J be the numbers of arrivals and departures on that side. Number the class events by their stable ranks. Let a count processed-side arrivals on even ranks and $o \in \{0, 1\}$ be the class's odd indicator. A class length m is count-feasible exactly when

$$m = \begin{cases} 2(I + J - 2a) + o, & t \text{ even}, \\ 2(J - I + 2a) - o, & t \text{ odd}, \end{cases} \quad m \ge 0, \quad 0 \le a \le \lceil m/2 \rceil, \quad 0 \le I - a \le \lfloor m/2 \rfloor. \tag{6}$$

Indeed the two rank parities provide precisely these available arrival slots; the local turn rule fixes departures. This is exact local count feasibility, not sufficient global chronological realizability.

For a frontier (b_0, b_1, c_0, c_1) of outgoing and incoming edge counts, the new row sees

$$(I, J) = \begin{cases} ((b_0, b_1), (c_1, c_0)), & t \text{ odd}, \\ ((b_1, b_0), (c_0, c_1)), & t \text{ even}. \end{cases}$$

If its lengths are m_0, m_1, the next frontier is

$$(m_0 - J_0, m_1 - J_1, m_0 - I_0, m_1 - I_1). \tag{7}$$

Start with $(1, 1, 1, 1)$ at a saturated rail and require the same frontier at the other rail.

Modulo four, the a term disappears from (6); hence the residue frontier is uniquely determined by the wake bits. It can be propagated in either direction. A wrong terminal residue rejects a mask at every period. A normalized physical event also satisfies

$$i \equiv 2r + (t \bmod 2) \pmod 4. \tag{8}$$

This follows directly by alternating vertical and horizontal unit moves. The four phase groups have $P/4$ events. Their count residues must agree, and their common residue fixes P modulo sixteen.

Proposition 5.1 (Finite input inventory)**.** *For* $h = -9, -7, -5, -3, -1$*, respectively, the modular tests leave* $12, 20, 12, 20, 12$ *normalized masks. Thus* 76 *wake assignments remain.*

Proof. Evaluate (6)–(7) modulo four for the 128 masks per profile, and check (8). The supplement regenerates every integer calculation in both directions and checks complementary masks. It also tests all 2304 masks before normalization and the profile parity rejection for the four even empty levels. No numerical optimizer is involved. □

For chronology pruning we also use nonnegative completion tables from (6)–(7). At each frontier retain the least cost of a counted completion. Because future count transitions depend only on the frontier, this is an exact minimum for the relaxation and a safe lower bound for a real completion. Tables are regenerated from the supplied wake; an incomplete user-provided table is not trusted as an exclusion.

6 Chronology-exact enumeration and its coverage

In each full class (t, r), list all events in stable order. Their turns are fixed: even ranks right, odd ranks left. Enumerate their incoming sides; literal turning fixes their outgoing sides. In a fixed checker sector an undirected lattice edge has one direction of use. Therefore its ordered source and target occurrences match first-to-first, with no free matching permutation. For class representatives $b_{t,r} = (r, t - r)$ the matching edge has integer voltage δ determined by

$$b_{\text{source}} + \text{outgoing unit vector} = b_{\text{target}} + \delta(2, -2).$$

Lemma 6.1 (Exact event-graph test)**.** *A fully joined event graph represents a finite-support periodic trace exactly when it forms one directed cycle with total voltage one and, along that cycle, the lifted times $j - Pv$ increase strictly with stable rank in every class.*

Proof. A real trace yields precisely such ordered edge matches and one phase cycle. Conversely the voltage identities reconstruct its lattice walk, headings and drift. The stated time order gives the alternating stabilized word of Lemma 2.1, which constructs a finite seed for the infinite repetition. A separate full-field parity calculation verifies every accepted seed. □

The searches retain all old sound pruning rules. Count tables only lower-bound unbuilt cost. A premature directed cycle cannot later join mandatory unbuilt events. The four phase groups cannot exceed the cap divided by four. Within a fixed open chain, consecutive reads of the same lifted cell must have consecutive stable ranks: a missing rank cannot be inserted into a trajectory segment whose every update is already fixed.

For completeness, the period-dependent time pruning does not assume the final period equals the cap. On an open chain write

$$\tau_u = z_{c(u)} + p_u - Pv_u.$$

Consecutive class ranks require a gap at least four by (8). A simple cycle of these difference inequalities on q chain variables has weight $C + KP$ with integer K and $C \geq 5q - n$, where n events have been built. If $K \geq 1$ it is positive at every $P \geq n$; if $K = 0$ its sign is constant; if $K < 0$ it decreases with P. Thus a positive constraint cycle at the search cap excludes every admissible smaller period. All operations use exact integers.

No pruning assumes bounded physical memory or a preselected seed box. Only completed traversals are exclusions. A time or node limit is explicitly inconclusive.

6.1 A disjoint partition of the final difficult case

One case, $h = -1$ and mask 187, requires a larger traversal. Its phase residues are all two, so $P \equiv 8 \pmod{16}$; the 192 upper bound therefore reduces to a search cap of 184.

The top-down search deterministically enumerates the 673 partial graphs at the completed-layer frontier preceding row -5. Shard j of eight follows exactly the ordinals congruent to j modulo eight. Each complete candidate crosses that frontier once. Since the upstream recursion and rollback are unchanged, all shards enumerate the same cut list, and their disjoint union covers the original search. Every shard must finish. Matching cut counts and digests are additional implementation controls, not a replacement for this partition argument.

7 Completed classification and proof of the main theorem

Table 1 gives the completed search totals. These count constructed prefixes, not distinct words or simulated ant updates. The last row includes repeated upstream prefixes in the eight shards.

Empty level	Period bound	Masks	Visited prefixes	Accepted traces
-9	192	12	56 388 171	none
-7	168	20	10 030 681	none
-5	160	12	2 353 673	none
-3	168	20	23 059 897	standard only
-1	192	12	299 233 450	none

Table 1: Completed exact enumeration. All caps are justified before the search; there is no period cutoff in the main theorem. The final difficult case uses cap 184 after its independently checked phase congruence.

All 76 canonical masks are covered by 83 completed native runs, counting the eight shards separately. The total is 391 065 872 prefix visits. The sole survivor is mask 163 with empty level -3. Its complementary mask 92 is the same geometric trace with the other top event chosen as origin.

The complete odd-class set is consequently forced to be

$$O_{\mathrm{std}} = \{(-10,0),(-10,1),(-9,1),(-8,1),(-7,0),(-6,0),\\ (-5,0),(-4,1),(-2,0),(-1,1),(0,0),(0,1)\}.$$

The normalized word, head east at the origin, is the concatenation of these two lines:

```
RRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLLRRRRLL
RRRRLLRLRRRRLRLLLLRRRRLRRLRRRRLLLLRLRRRRLRRRRLLLLRLR
```

It is a cyclic phase of the fixed standard word. A complete seed is

$$B = \{(-7,0),(-7,1),(-6,-3),(-6,-2),(-6,-1),(-6,2),\\ (-5,-2),(-5,0),(-5,1),(-5,2),(-4,-2),(-4,1),\\ (-3,-3),(-3,1),(-2,-3),(-2,-1),(-2,1),(-1,-2),(-1,0)\}.$$

Nineteen is not claimed to be the minimum seed size over other phases.

Proof of Theorem 1.2. Proposition 3.1 gives the five profiles and saturated rails. Complementary-origin normalization and Proposition 5.1 leave the 76 masks. Lemma 4.1 bounds every possible period,

and the chronology-exact enumeration covers every candidate below these bounds by Lemma 6.1 and the sound pruning arguments. The completed inventory contains exactly the displayed standard word. Its separate infinite-future verification supplies existence. Therefore $P = 104$ and the wake is standard. □

8 Evidence, reproducibility and remaining questions

The core theorem is computer-assisted: its logical coverage is a written mathematical argument and its finite claims are checked by integer programs. The supplemental source hashes identify the executed programs and the complete run records identify the finite exclusions.

The supplement contains the unchanged inherited count routines, the two ordered-event traversal directions, the local rank check, the upper-bound recipe, all parameterized sources actually executed, and the complete native records. A regenerated count implementation is cross-checked with the retained one. The small shard controls compare their union with an unsplit positive search. The full replay checks every case, all eight final shards, and the unique finite seed. Single active-cell deletions and a far-ahead blocker are rejected by the independent infinite verifier.

```
python3 wake_synthesis/reproduce.py --jobs 4
python3 wake_synthesis/reproduce.py --quick
```

The second command explicitly skips the 76-case exclusion replay, while checking its upper certificate, count tables and controls. The default is full replay. Python's standard library and a C++17 compiler are sufficient; no linear optimizer, SAT solver or network access is required. Resource-limited exploratory runs are retained as such and are not used as negative evidence.

The theorem isolates a genuinely narrower residual periodic problem: a nonstandard highway retaining drift $(2, -2)$, growth twelve and width ten must have a thicker interior wake. It remains necessary to determine whether such wakes exist, whether different numerical invariants are possible, and whether all finite seeds eventually become periodic. Those questions are not answered by the present classification.

A Wake-dependent period bounds and inverse synthesis

The standard-wake classification is one instance of a general inverse problem. Prescribe a non-axis drift d, an initial heading, and a finite set $O \subset \mathbb{Z}^2/\mathbb{Z}d$ of odd classes. We seek a periodic word and finite seed with exactly these data. The following bounds, developed in the first version [5], determine a finite search range directly from the input wake.

Lemma 2.1 gives $\#\mathrm{R} - \#\mathrm{L} = |O| =: g$. Heading restoration implies $g \equiv 0 \pmod 4$, P even and $d_x + d_y$ even. For $d = (m, -m)$, the complete class label is $(x + y, x \bmod m)$. The checker-sector invariant σ is determined by the supplied pose.

A.1 An integer clock for diagonal budgets

After rotating a diagonal drift parallel to $(1, -1)$, the transverse coordinate $x+y$ and heading return after each period. A clock using these quantities has a cancelling head correction: its period gain comes from the wake. We recall the unit increment identity of [4] before computing that gain.

Fix an integer c, set $s = x + y - c$, and put

$$f(s) = 3^{|s|} - 1, \qquad \epsilon(h) = \begin{cases} +1, & h = N, E, \\ -1, & h = S, W, \end{cases} \qquad G(s, e) = \begin{cases} \frac{1}{2} f(s), & es \geq 0, \\ -\frac{3}{2} f(s), & es < 0. \end{cases}$$

Use the integer clock of [4] on the complete finite state

$$\Theta_c(X) = -\sigma(X) \sum_{q \in B} \chi(q) f(q_x + q_y - c) + G(p_x + p_y - c, \epsilon(h)). \tag{9}$$

Since $f(s)$ is even, Θ_c is an exactly computable integer.

Lemma A.1 (Integer clock, Ke [4])**.** *For every finite-support state and integer centre c, with e, e' the before and after values of ϵ and $s = p_x + p_y - c$,*

$$\Theta_c(FX) - \Theta_c(X) = 1 + 4f(s)\mathbf{1}_{\{e' \neq e,\ es<0\}} \ \geq 1. \tag{10}$$

Proof. Write $\delta = +1$ on R and -1 on L. Directly checking the two colours at four headings gives $\delta\sigma\chi(p) = ee'$, and the outgoing step gives $s' = s + e'$. Thus the black-field increment is $-ee'f(s)$. Let $G_\pm(s) = G(s, \pm 1)$. The elementary recurrences are

$$G_+(s+1) - G_+(s) = f(s) + 1, \qquad G_-(s-1) - G_-(s) = f(s) + 1,$$
$$G_-(s) - G_+(s) = -2\,\mathrm{sgn}(s) f(s).$$

If $e' = e$ the total gain is one. For $+1 \to -1$ it is $1 + 2f(s) + G_-(s) - G_+(s)$, equal to one for $s \geq 0$ and to $1 + 4f(s)$ for $s < 0$. The other reversal gives the reflected cases. This includes $s = 0$. □

A.2 Diagonal wake budgets and crossing refinement

Rotate a diagonal drift to $d = (m, -m)$, $m > 0$, and use the transformed heading. Let $t = x + y$, let o_t count odd classes at level t, and define

$$A_t = -\sigma(-1)^t o_t, \qquad K_c(O) = \sum_t A_t(3^{|t-c|} - 1). \tag{11}$$

Theorem A.2 (Wake-determined diagonal period)**.** *Every realization of this wake obeys $P \leq K_c(O)$ for every integer c.*

Proof. Over one period the head term of (9) cancels, since both t and the heading return to their initial values. The signed cell toggles give $-\sigma \sum_i \chi(q_i) f(t_i - c)\delta_i$. The weight is constant in each full d-class. Lemma 2.1 makes its net δ count one on odd classes and zero on even classes. The exact period increment is therefore $K_c(O)$. Summing (10) proves the bound. The unchanged part of the full field cancels algebraically; it is not deleted. □

Here is an independent geometric derivation, which also supplies a refinement. The finite projected period is a closed walk with increments ± 1 in t. Let f_t count upward crossings of the cut $(t, t+1)$; it also counts downward crossings. Let m_t count visits to t, and u_t count passages entering and leaving in the increasing direction. Flow conservation equates this with the number of decreasing-through passages. The fixed arrival axis and the two turn choices give

$$m_t = f_{t-1} + f_t = A_t + 4u_t, \qquad 0 \leq u_t \leq \min(f_{t-1}, f_t). \tag{12}$$

For example, at vertical-arrival levels, right turns are through passages and left turns reverse the projection, giving $\#\mathrm{R} - \#\mathrm{L} = 4u_t - m_t$; horizontal arrival reverses that sign. Substituting $u_t \leq f_{t-1}$ into $f_t = A_t + 4u_t - f_{t-1}$ gives the first inequality below; $u_t \leq f_t$ gives the second:

$$f_t \leq 3f_{t-1} + A_t, \qquad f_{t-1} \leq 3f_t + A_t. \tag{13}$$

Let ℓ, h be the extreme wake levels. Outside them $A_t = 0$; starting past the finite support of the projected walk and applying (13) inward forces $f_{\ell-1} = f_h = 0$. Thus the periodic trace lies between the extreme wake levels. An empty wake forces every $f_t = 0$, impossible for a nonempty period.

Starting from these two zero boundary values, set

$$L_{\ell-1} = 0, \quad L_t = 3L_{t-1} + A_t, \qquad R_h = 0, \quad R_t = 3R_{t+1} + A_{t+1}, \quad C_t = \min(L_t, R_t).$$

Then $0 \le f_t \le C_t$ for $\ell \le t < h$. If a C_t is negative, the wake is impossible; otherwise

$$P = 2\sum_t f_t \le 2\sum_t C_t. \tag{14}$$

Moreover $d_x = \sigma \sum_t (-1)^t f_t$: on a cut with $\sigma(-1)^t = 1$, the upward horizontal moves contribute $+f_t$ to x; on the other cuts the downward horizontal moves contribute $-f_t$. If C_+ and C_- sum the corresponding positive and negative cut bounds, then also

$$P \le 4C_+ - 2d_x, \qquad P \le 4C_- + 2d_x. \tag{15}$$

Corollary A.3 (Width and growth bound)**.** *Writing $W = h - \ell$, a diagonal realization has even W and*

$$P \le g(3^{W/2} - 1). \tag{16}$$

Proof. At a transverse extreme every passage reverses the projection. The fixed arrival axis makes all letters in an extremal class equal. Lemma 2.1 therefore forces exactly one R per such class. Both extremes have the same arrival axis, so W is even; this is the usual extremal rigidity argument. Take $c = (\ell + h)/2$ in Theorem A.2, discard negative coefficients and use $\sum_t \max(A_t, 0) \le g$ to obtain (16). □

For the normalized standard wake, $\sigma = -1$ and the coefficients at $t = -10, \dots, 0$ are

$$A = (2, -1, 1, -1, 1, -1, 1, 0, 1, -1, 2).$$

The cut bounds are $C = (2, 5, 16, 47, 142, 145, 48, 16, 5, 2)$, with sum 428. Thus (14) gives $P \le 856$, as does (11) with centre $c = -5$. The chronological constraints in Section 4 strengthen this budget to the cutoff used in the complete classification.

A.3 General non-axis drift by reciprocal moments

For a general drift, use a transverse linear coordinate constant along it. An exponential weight and its reciprocal give complementary period budgets; Cauchy–Schwarz then bounds the number of steps independently of their individual weights.

Rotate the drift to $(a_0, -b_0)$, $a_0, b_0 > 0$. Put $k = \gcd(a_0, b_0)$, $\eta = (b_0/k, a_0/k)$, and choose rational $r > 1$ such that, with $a = r^{\eta_x}$ and $b = r^{\eta_y}$,

$$\kappa = 1 - \frac{2}{a+1} - \frac{2}{b+1} > 0.$$

The choice $r = 4$ always works. Define $w(q) = r^{\eta \cdot q}$, and

$$(\gamma_N, \gamma_E, \gamma_S, \gamma_W) = \left(\frac{2}{b+1}, \frac{2}{a+1}, -\frac{2b}{b+1}, -\frac{2a}{a+1} \right).$$

The one-sided potentials are

$$\Psi_+(X) = -\sigma \sum_{q\in B} \chi(q)w(q) + \gamma_h w(p), \qquad \Psi_-(X) = -\sigma \sum_{q\in B} \frac{\chi(q)}{w(q)} + \frac{\gamma_{h+2}}{w(p)}.$$

Substitution of the current-cell flip gives $\Delta\Psi_+ = \kappa w(p)$ in six heading/colour cases and $(\kappa+4)w(p)$ in S/L and W/R. The half-turn formula gives $\Delta\Psi_- \geq \kappa/w(p)$. These local identities are the one-sided exponential clock construction, included to make the new period argument self-contained.

Define the two finite wake moments, using any representatives of O,

$$A_+ = -\sigma \sum_{q\in O} \chi(q) r^{\eta\cdot q}, \qquad A_- = -\sigma \sum_{q\in O} \chi(q) r^{-\eta\cdot q}.$$

They are independent of the representatives since $\eta \cdot d = 0$ and $d_x + d_y$ is even. The head terms cancel over a period, and Lemma 2.1 gives exact increments A_+ and A_-. Thus both are positive and $\kappa \sum_i w(q_i) \leq A_+$, $\kappa \sum_i w(q_i)^{-1} \leq A_-$. Cauchy–Schwarz yields:

Theorem A.4 (Reciprocal wake bound)**.** *Every realization with the specified non-axis drift and wake satisfies*

$$P^2 \leq \frac{A_+ A_-}{\kappa^2}. \tag{17}$$

All quantities are rational. The integer square root of the floor of the right side is a computable cutoff. Changing the transverse centre scales A_+ and A_- reciprocally, leaving the product unchanged. For axis drift the normal has a zero coordinate, so this construction cannot give positive κ.

A.4 All-period synthesis and detour obstruction

Theorem A.5 (Period-unknown inverse synthesis)**.** *Given a non-axis integer drift, an initial heading and a finite odd-class wake, existence of a finite-support periodic realization is decidable. A positive answer can produce its word and an explicit finite seed.*

Proof. First reject odd $d_x + d_y$, growth $|O|$ not divisible by four, or a nonpositive moment A_+ or A_-. Otherwise compute Theorem A.4's cutoff, additionally using Theorem A.2 and (14)–(15) in the diagonal case. A negative crossing bound also rejects the input. Enumerate every nonempty word up to the least derived bound. Check its heading and drift, apply Lemma 2.1, and require the exact odd-class set. A passing word constructs a finite seed; complete finite exhaustion gives a negative answer. Completeness follows from the period bound. □

The all-future verifier of Section 8 independently checks the constructed finite seed. For cutoff B, enumeration visits at most $2^{B+1} - 2$ nonempty binary words; the period bound guarantees termination.

Corollary A.6 (No wake-preserving closed-pose insertion)**.** *Suppose realizable words u, v have the same non-axis drift, and v is obtained by inserting nonempty detours into u, each returning to the same position and heading while leaving the original steps unchanged. Then*

$$A_+(O_v) > A_+(O_u), \qquad A_-(O_v) > A_-(O_u).$$

Thus their signed transverse wake cannot be identical. For diagonal drift, $K_c(O_v) - K_c(O_u) \geq |v| - |u| > 0$ for every integer c.

Proof. Every retained original step has the same position, heading and letter; hence it has the same local clock increment. Every inserted step has strictly positive increment in both one-sided clocks, and at least one in the integer clock. Sum over the words and use the exact wake-increment equalities. □

Thus any such realizable detour requires an additional wake budget.

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